

Zero-Knowledge Proofs

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Lecture notes

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Definition 1.1 (Alphabet).

$$\Sigma := \{0, 1\}.$$

Definition 1.2 (Statement).

$$x \in \Sigma^*.$$

Definition 1.3 (Language).

$$\mathcal{L} \subseteq \Sigma^*.$$

Definition 1.4 (Complexity class).

$$\{\mathcal{L}\}.$$

Definition 1.5 (Deterministic Turing machine).

$$\begin{aligned} \delta : Q \times \Gamma &\rightarrow Q \times \Gamma \times \{L, R\}, \\ |Q| < \infty, \quad \Gamma \supseteq \Sigma, \quad \sqcup \in \Gamma \setminus \Sigma, \quad q_0, q_{\text{acc}}, q_{\text{rej}} \in Q, \\ M &:= (Q, \Sigma, \Gamma, \delta, q_0, q_{\text{acc}}, q_{\text{rej}}). \end{aligned}$$

Definition 1.6 (Decision algorithm, efficiency). A *decision algorithm* is a DTM $M : \Sigma^* \rightarrow \{0, 1\}$ that halts on every input and *decides* $\mathcal{L}$ if

$$\forall x \in \Sigma^*, \quad M(x) = 1 \iff x \in \mathcal{L}.$$

M is *efficient* (polynomial time) if

$$\exists q \in \text{poly} : \quad \forall x \in \Sigma^*, \quad \text{time}_M(x) \leq q(|x|).$$

Definition 1.7 (P).

$$\mathbf{P} := \left\{ \mathcal{L} \mid \exists M, q \quad \forall x \in \Sigma^*, \quad M(x) = 1 \iff x \in \mathcal{L}, \quad \text{time}_M(x) \leq q(|x|) \right\}.$$

Definition 1.8 (NP).

$$\text{NP} := \left\{ \mathcal{L} \mid \exists \mathcal{R}_{\mathcal{L}}, q : \forall x \in \mathcal{L} \exists w : |w| \leq q(|x|), (x, w) \in \mathcal{R}_{\mathcal{L}} \right. \\ \left. \wedge \forall x \notin \mathcal{L} \nexists w : (x, w) \in \mathcal{R}_{\mathcal{L}} \wedge \mathcal{R}_{\mathcal{L}} \in \mathcal{P} \right\}.$$

w is the *witness* of x .

Remark 1.9 (Traditional proofs).

$$\mathcal{L} \in \mathcal{P} \Rightarrow x \in \mathcal{L} \text{ needs no proof; } \quad \mathcal{L} \in \text{NP} \Rightarrow x \in \mathcal{L} \iff \exists w : (x, w) \in \mathcal{R}_{\mathcal{L}}.$$

Definition 1.10 (Interactive algorithms). $P, V : \Sigma^* \rightarrow \Sigma^*, k : \Sigma^* \rightarrow \mathbb{N}$, coins r, s :

$$a_1 = P(x, r), \quad a_2 = V(x, a_1, s), \quad a_3 = P(x, a_1, a_2, r), \dots \\ \langle P(r), V(s) \rangle(x) = a_{k(x)}, \quad \text{transcript} = (x, a_1, \dots, a_{k(x)}).$$

Definition 1.11 (IP). $\mathcal{L} \in \text{IP} \iff \exists P, V$ with V efficient ($\text{poly}(|x|)$) and

$$\underbrace{\forall x \in \mathcal{L} : \Pr_{r,s}[\langle P(r), V(s) \rangle(x) = 1] \geq \frac{3}{4}}_{\text{completeness}} \wedge \underbrace{\forall x \notin \mathcal{L}, \forall P^* : \Pr_{r,s}[\langle P^*(r), V(s) \rangle(x) = 1] \leq \frac{1}{2}}_{\text{soundness}}.$$

(P, V) is an interactive proof system for $\mathcal{L}$; P may be unbounded.

Definition 1.12 (Graph isomorphism).

$$G_0 \cong G_1 \iff \exists \pi \in \text{Perm}(n) : G_0 = \pi(G_1), \\ \mathcal{L}_{GNI} := \{(G_0, G_1) : G_0, G_1 \subseteq [n]^2, G_0 \not\cong G_1\}.$$

Protocol $\mathcal{L}_{GNI}$

$$\begin{array}{ll} P(G_0, G_1) & V(G_0, G_1) \\ & b \xleftarrow{\$} \{0, 1\}, \quad \pi \xleftarrow{\$} \text{Perm}(n), \quad H := \pi(G_b) \\ & \xrightarrow{H} \\ B : H \cong G_B \text{ (try all perms)} & \xleftarrow{B} \text{ output } (B = b) \end{array}$$

Analysis. Completeness ($G_0 \not\cong G_1$): $H \cong G_0 \wedge H \cong G_1 \Rightarrow G_0 \cong G_1 \#$, so H gives a unique $B = b$. Soundness ($G_0 \cong G_1$): $\pi(G_0) \sim \pi(G_1) \Rightarrow B \perp b \Rightarrow \Pr[B = b] = \frac{1}{2}$.

Definition 1.13 (Algorithmic knowledge).

$$\text{knowledge} := \{\text{anything an algorithm computes efficiently}\}.$$

Definition 1.14 (Perfect zero-knowledge).

$$\text{View}_V^P := (x, s, a_1, \dots, a_{k(x)}),$$

$$(P, V) \text{ perfect ZK} \iff \forall V^* \exists S : \forall x \in \mathcal{L}, \text{View}_{V^*}^P = S(V^*, x).$$

V^* = malicious verifier; S = efficient simulator without a witness.

Definition 1.15 (Graph isomorphism).

$$\mathcal{L}_{GI} := \{(G_0, G_1) : G_0 \cong G_1\}, \quad \mathcal{R}_{GI} := \{(G_0, G_1, \pi) : G_0 = \pi(G_1)\} \in \mathbf{P} \Rightarrow \mathcal{L}_{GI} \in \mathbf{NP}.$$

Protocol $\mathcal{L}_{GI}$, perfect ZK

$$\begin{array}{ll} P(G_0, G_1) & V(G_0, G_1) \\ \sigma \xleftarrow{\$} \text{Perm}(n), \quad H := \sigma(G_0) & \\ & \xleftarrow{H} \\ \tau := \begin{cases} \sigma, & b = 0 \\ \sigma \circ \pi, & b = 1 \end{cases} & b \xleftarrow{\$} \{0, 1\} \\ & \xrightarrow{\tau} (H = \tau(G_b)) \end{array}$$

Analysis. Completeness: $b = 0 \Rightarrow \tau(G_0) = H$, $b = 1 \Rightarrow \tau(G_1) = \sigma(\pi(G_1)) = H$. Soundness ($G_0 \not\cong G_1$): $G_b \cong H$ for ≤ 1 of $b \in \{0, 1\} \Rightarrow \Pr_b[\text{accept}] \leq \frac{1}{2}$.

Protocol $S(V^*, G_0, G_1)$

$$\begin{array}{ll} \tau \xleftarrow{\$} \text{Perm}(n), \quad B \xleftarrow{\$} \{0, 1\}, \quad H := \tau(G_B) & \\ & b \leftarrow V^*(G_0, G_1, H) \\ b \neq B \Rightarrow \text{restart} & \\ \text{output } (G_0, G_1, \tau, H, b) & \end{array}$$

Why valid. S guesses b backwards; efficient since V^* is and the loop clears in $\mathbb{E} = 2$ tries; $G_0 \cong G_1 \Rightarrow \tau(G_0) \sim \tau(G_1)$, so (τ, H, b) matches the real view.

Efficiency targets

	P -time / proof	
intro. (defs)	poly(N), poly(N)	(not ZK)
sigma protocols	poly(N), poly(N)	ZK
short arguments	poly(N), polylog(N)	ZK
non-interactive	$O(1)$, $O(1)$	ZK